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Designing Generative Artificial Intelligence Integrated Immersive Virtual Reality Therapeutic Experiences to Support Meaning-centered Grief Therapies for Bereaved Parents

Lei Zhang*
Kennesaw State University

Anisah Bagasra
Kennesaw State University

Erica Kaye
St Jude Children's Research Hospital

Wendy G. Lichtenthal
University of Miami

Abstract

Bereaved parents often struggle to make sense of and find meaning in and after their loss due to the untimely nature of their child's death. Challenges with finding meaning have been associated with prolonged grief symptoms that can adversely affect grieving parents' quality of life. Recently, meaning-centered grief interventions that focus on facilitating adaptive meaning-making after loss have shown therapeutic benefits in reducing prolonged grief symptoms in bereaved individuals. However, these interventions are mostly administered in one-on-one, in-person settings, which limits their accessibility to a broader population due to logistical and locational constraints. While existing videoconferencing applications can provide remote bereavement support and deliver grief therapies, they often lack engaging user experiences and features to support high interactivity, guided imagery and mindfulness exercises commonly used in meaning-centered grief therapies. We hypothesize that the immersive and interactive nature of a networked immersive virtual reality (IVR) system, enhanced with custom content creation features from text-to-image generative artificial intelligence (genAI) models, will create a unique and individualized therapeutic experience that supports meaning-centered grief techniques and activities for bereaved parents in a shared virtual world. This work-in-progress paper presents the design concept of the system prototype, preliminary survey results on user acceptance of the technology, and selected user experience scenarios.

1 Introduction

It is argued that people are driven by a need to find or create a sense of purpose, or meaning, in their lives. Therefore, the distress caused by the loss of a significant other often leads to a quest for meaning in the loss by bereaved individuals [1]. Searching for meaning comes as a central psychological struggle for bereaved parents because the tragic and untimely loss of a child creates difficulty fitting the experience into a pre-existing worldview and results in a sense of injustice in life. Inability to find meaning and effectively integrate the loss into one's worldview is associated with more intense and lasting symptoms of grief in bereaved parents [2, 3], putting them at high risks of developing prolonged and complicated grief (a state of grief in which grief symptoms interfere with an individual's ability to function in daily life). Statistics show that about 10-20% of traumatic grief develops into complicated and prolonged grief and affects grievers' quality of life [4, 5, 6].

*e-mail: lzhang24@kennesaw.edu

The recent concept of meaning-making and reconstruction as a key role in effectively processing grief emerges as a new grief model that leads to the designs of new grief interventions for bereaved individuals. Specifically, meaning-centered psychotherapies for bereavement have seen positive evidence in application to complicated grief [7, 8]. Although studies of meaning-centered psychotherapies have been conducted in both in-person and remote formats through one-on-one sessions, each format has its own drawbacks. The in-person therapy provides the best communication and interactive experience for doing meaning-centered grief activities but has limited access to a wider population of bereaved individuals due to constraints of location and logistics. In comparison, remote grief therapies are normally conducted via videoconferencing applications (e.g. Zoom, Webex, etc.) that overcome accessibility constraint, but with very limited built-in tools to support meaning-centered grief activities, such as creating artwork to memorialize loved ones, custom imagery and visualization exercises for mindfulness and meditation, etc. Additionally, the computer screens present an invisible barrier (the "fourth wall effect") between bereaved parents and therapists, affecting the buildup of empathy and intimacy between them when they share sensitive grief stories with each other. Therefore, there is a strong, unmet need to develop networked, immersive, interactive, and individualized digital therapeutic experiences that simulate the real-world experiences of meaning-centered grief interventions.

On the other hand, the current advancement of immersive technologies and genAI and their increasing affordability and accessibility make it possible for human-computer interaction (HCI) researchers to create new research testbeds to investigate how the grieving experiences and the meaning-making and reconstruction process can be supported in bereaved individuals through rich affordances by the integration of both technologies, such as, immersion, sense of "being there", interactivity, and custom automatic content generation. Currently, research in this area is very lacking due to the sensitivity of the topic, and with a rapid increase and adaptation of immersive technologies in virtual bereavement and grief services since the pandemic, it is crucial to investigate how system designs may affect grieving process in a technology mediated immersive virtual world. With such rationales, we propose the design of a networked IVR system with integration of text-to-image genAI models to support specific meaning-centered grief activities for bereaved parents.

2 Related Work

2.1. Meaning-making and Reconstruction In Grief Therapies

The field of grief process research has seen a shift from traditional stage-based theories to more postmodern or constructivist theories in the past few decades [1, 9]. The most recent constructivist grief theories emphasize the important role of meaning making and

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reconstruction during the grief process. Flesner [7] defines three essential components within the concept of meaning making as, sense-making (the ability of the person experiencing the loss to understand it), benefit finding (the ability to recognize factors that are positive from the loss), and identity change (the positive or negative reconstruction of self) and suggests that these components are navigators that can help bereaved individuals work through grief. When applied to grief therapies, meaning-making has been integrated into several grief techniques. One is narrative reconstruction through story sharing or creating new stories to make meaning from the loss [10]. Another one is narrative art therapy through different forms such as painting, photography, drawing, or crafts (Neimeyer, 2014). In a case study, Neimeyer and Thompson [8] integrated expressive drawing activities for meaning making into an individual grief therapy with a patient of complicated grief and the results showed that the activity relaxed the patient from symptoms and connected her more with her dead mother and she was finally able to manage her daily life.

2.2 VR Supported Grief Therapies

To date, IVR has been considered as an efficacious tool in many psychotherapy studies to support treatments of anxiety, phobias, and post-trauma stress disorder (PTSD) [11, 12, 13]. However, studies using IVR in grief therapies are very limited in literature. *Emma's world* [14] is an adaptive display and VR system developed as an exposure-based treatment to reduce anxiety disorders from complicated grief. The system includes components such as education, slow breathing training, exposure, cognitive restructuring, etc. Users have the option to choose one of the four landscape environments based on their present mood and interact with a digital book, *Book of Life*. The adaptability of the system shows the potential of a VR system to adjust to the needs of the users. In the study, a female patient was treated and results showed significant reduction of complicated grief related symptoms and maintained effect at a 12-month follow up. Knowles, et al. [15] conducted a pilot study on the feasibility and acceptability of using a social game, *Second Life*, to support grief in widowers. Compared with website based virtual grief support, the 8-week desktop VR based support showed significant improvement in depression related to grief. This study shows the efficacy of a networked, multi-user, desktop VR system in treating grief.

These existing studies of using VR in grief therapies all achieved promising results on the effectiveness of the system in reducing grief-related symptoms. However, none of the systems were fully immersive and AI-enhanced, and the user's ability to customize the scene based on personal grief experience is limited. Additionally, the studies were not designed to specifically focus on application of meaning-making process in VR either. Therefore, the proposed networked, IVR system with GenAI integration will be a huge improvement in its design compared to these old VR systems.

2.3 Integration of GenAI in IVR

The power of GenAI to enhance user experiences, automate content creation, and enable intelligent interactions has given it an indispensable role in generating virtual content for a dynamic virtual world [16]. The idea of integrating GenAI into IVR presents a fairly new research topic and currently limited studies have focused on both the conceptual framework and actual applications. Ratican et al. [17] proposed a framework for 3D asset generation using GenAI in Metaverse, in which a development pipeline of using text-to-image and image-to-3D generative models was discussed. Chheang et al. [18] presented a study on using a generative virtual assistant in VR anatomy education by integrating text-to-speech model and ChatGPT into an avatar in IVR to guide a virtual anatomy class and compared it to a screen-based representation. Study results showed that the AI assistant in the form of either an avatar or a screen-based agent performed equally well in assisting medical students with their learning. This study demonstrated the feasibility and usability of integrating a GenAI model into the IVR system.

3 Project Prototype Design

We started the project by reaching out to the Quality of Life (QoL) and Bereavement program at a major pediatric cancer treatment hospital and connected with program coordinator and clinical researchers there. With their assistance, we were able to join their bereaved parent steering council and an online community, *LegacyVoices*, for all bereaved parents from the hospital and get in touch with them. Meanwhile, we were also able to connect to a clinical researcher who is currently conducting clinical trials of remote meaning-centered grief therapy with bereaved parents from cancer and get her perspectives on simulating some of the therapy sessions in our envisioned IVR and genAI integrated system. We focused on the bereaved parents who have lost a child to cancer specifically in this project because cancer is the leading cause of death by disease in children [19]. And as caregivers, bereaved parents have been living with extreme stress ever since the cancer diagnosis in their children and are at high risk of mental breakdown and mental disorders.

3.1 A User Acceptance Survey

Based on our conversations with grief researchers and clinical psychologists, and due to the sensitive nature of the research project, our first step is to do a user need assessment survey (Appendix. I) with the bereaved parents to seek their attitudes on using IVR for bereavement support and their willingness to use the IVR prototype afterwards. After obtaining IRB permission, we distributed the survey to the mailing list of *LegacyVoices* and a physical bereavement camp and collected 27 effective responses. The survey result shows that, overall, a total of 78% to 81% of participants (Very open, 48% and somewhat open, 30%-33%) indicated that they are open to the idea of using VR for grief therapy (78%) and bereavement support (81%). Regarding using VR for virtual memorial services or bereavement events, 48% of participants indicated *Yes* and 37% indicated *Maybe*. Lastly, when asked how likely they are willing to try the VR grief prototype when it's available, 64% of participants (Very likely, 39%; likely 25%) indicated they would be likely to try it. In conclusion, we found a majority of the bereaved parents who participated in the survey were open to the idea of using VR for grief and bereavement support and there is a high likelihood that they will try the project prototype in future.

3.2 Design Inspirations

Our proposed project prototype design was inspired by various grief and bereavement events, activities and clinical interventions, which mainly include:

1. Grief activities from bereavement events (e.g. Camp Sunshine bereavement camp, Day of Remembrance) organized by collaborating QoL and Bereavement program.
2. Sessions from a *Meaning-centered Grief Therapy* [20] currently in a clinical study.
3. Perspectives from user acceptance survey and collaborating clinical and grief researchers

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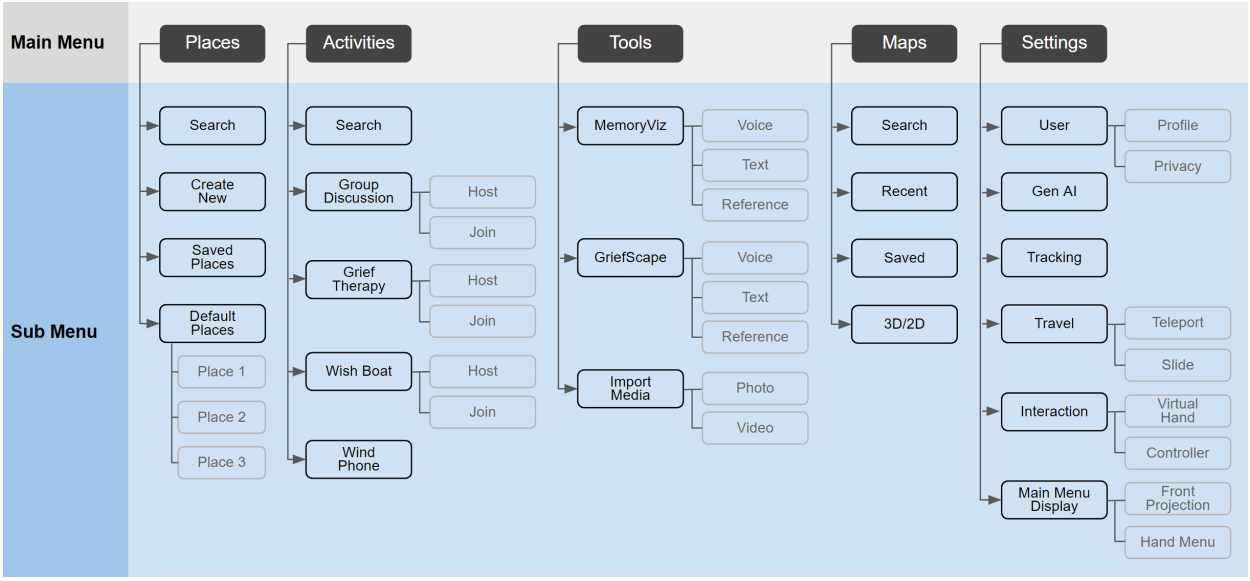

Figure 1. Prototype information architecture with the proposed system features

Specifically, at both bereavement camp and Day of Remembrance, a parent group discussion session provides participating bereaved families an opportunity to share their grieving experiences and the stories of the passed children, which creates a healing moment for the bereaved parents through similar experiences and helps build up strong empathy and foster connectedness between them. Wish boat ceremony and memory board creation are another two meaning-centered activities in these two events to help the bereaved parents to build up a continuous bond with their deceased children. Most parents enjoy the process of creating custom artworks in these two activities to celebrate the life of their children.

The *Meaning-centered Grief Therapy* [20] contains 16 sessions of meaning-making and reconstruction focused conversations between a bereaved parent and a therapist. Some major sessions of the therapy engage the parents in building a continuing bond between them and their children by having them share both theirs and their children's stories. Expressing emotion through a mindfulness guided imagery exercise is another key component in the sessions. A final Living Legacy Project culminates the therapy with parents' involvement of artistic and creative work to honor their children.

We held six group meetings with collaborating grief researchers and clinical psychologists to discuss the scope of the project, risks, and envisioned project prototype features that will be useful in clinical settings. We created detailed user experience scenarios to get feedback from these experts. Information from all these meetings and experts' opinions contributed to our design decisions enormously.

3.3 Design of the Project Prototype

Based on the design inspirations and our research of commonly used meaning-centered grief therapy techniques, we have decided that the project prototype needs to support some core meaning-centered grief activities, such as, a group story sharing session, custom imagery creation for mindfulness, art creation and object construction. We specified the project prototype with the following system features to support these grief activities (Figure 1).

- Support multi-user group activities, such as group discussions and collaborative art creation.
- Support user sharing of media from personal mobile devices in VR.

  This feature allows a parent to share his/her child's photos and videos to a virtual screen in VR when telling the child's stories in a group discussion session.
- Support custom environmental imagery generation (360 skybox) in VR using text-to-image genAI.

  This feature allows a parent to visualize his/her emotions/grief status to others using a Skybox AI environment generation tool in VR. It can also allow the parent to create a virtual place that resembles the one from one's past memory.
- Support object construction and art creation with rich virtual building tools and an intuitive UI.

  This feature allows the parents to build custom wish boats for virtual rituals and ceremonies.
- Support a Wind Phone scenario

  This feature allows the parents to talk to their deceased children in an unconnected phone in the virtual world and build an invisible bond to them.

3.4 User Experience Scenarios

To help our collaborators and the bereaved parents better understand and envision how the proposed project prototype works, we created a series of user experience scenarios based on specific system features. The scenarios include envisioned user experiences from the roles of both a bereaved parent and a clinician who are using the system and covers following grief activities: 1. A group discussion session (group therapy) 2. A one-on-one grief therapy session 3. A wish boat creation and ceremony session 4. A wind phone session. Due to paper length constraint, we present 2 selected scenarios below. Scenario 1 envisions a group therapeutic experience in which a bereaved father shares his daughter's stories with others. Scenario 2 demonstrates how a bereaved mother uses an integrated genAI tool to create a custom virtual environment from a past memory and use it for her individual grief therapy session in VR.

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3.4.1 Scenario 1. A bereaved father shares his child's story in a group discussion session in VR

John is a 41-year-old bereaved dad who lost his 2-year-old daughter, Lexie, to leukemia about 6 months ago. He feels very isolated from his social network because he doesn't know how to share the traumatic experience with others and is concerned if they understand his feelings. One day, he received an invitation from his daughter's treatment hospital to participate in a remote bereaved parent discussion group in virtual reality, which sounds novel to him. So he confirmed his participation and received a Meta Quest Pro VR headset with two controllers in a few days.

Figure 2. A concept illustration of VR group discussion therapy

John follows the instructions to set up the VR headset at home and connects it to *DigiPresence*, a virtual world specially designated to bereaved parents who have lost a child to cancer. A virtual assistant greets him and directs him to choose an avatar for himself with customization options. After that, John is asked to select *Activities* > *Group Discussion*> *Join* to enter the virtual space for group discussion. After selection, the surrounding environment fades out and fades in in a few seconds and the scene changes. Looking around, John finds himself sitting with other virtual avatars in a circle by the beach, with lighthouses and a beautiful sunset in the distance (Figure 2). An avatar sitting next to him greets him. John notices a name tag hovering over her head blinking when she speaks. Around the meeting time, a female avatar, called Sandra from her name tag, who seems to be a discussion group moderator and lead, introduces herself to the group and asks everyone to take turns sharing stories of their child.

John is a bit nervous about how to talk about his daughter at first, but soon he feels relaxed because the stories and feelings other parents have shared are so familiar to him and he really wants to share his own with the others. John notices that a big screen hovers in mid-air and can play both photo slides and videos of the child that a bereaved parent wants to share with others. When John talks about his daughter, Lexie, he really wants to share a short video of her when she was happy and healthy. He opens the *Tools* icon hanging around his left virtual hand and chooses *Import Media/Video*. A dialog box pops up and asks him to select a device to share from. He selects his smartphone and waits for a few seconds for it to get connected. After allowing permission to access the phone gallery, John is able to browse and select a video from favorites. Soon the video is projected to the virtual screen in VR and plays for everyone. The one-hour meeting flies by quickly as parents have a lot to talk about their children with media sharing. John also feels relaxed after sharing a lot of lovely memories of his daughter with the group. He begins to understand the meaning of sharing Lexie's stories to others. That is, by constantly mentioning and involving her in his daily life activities, the connection between Lexie and him continues even though she is not physically there.

3.4.2 Scenario 2. A bereaved mother creates a custom virtual environment for her one-on-one grief therapy

Lisa is a 35-year-old bereaved mom who lost her 8-year-old son Trevor about two years ago. She has developed some prolonged grief symptoms since then and has been working with a grief therapist, Sandra, remotely to improve her symptoms. A few weeks ago, Sandra invited her to a VR grief therapy session, and though this was new to her, she accepted.

When the session begins, Lisa puts on her VR headset and follows the instructions to load the virtual world, *DigiPresence*. Then, she selects *Activities* > *Grief Therapy* > *Join*. Immediately, she finds herself in a small cozy cabin room, and Sandra's avatar sits next to her. Seeing Lisa's avatar has arrived, Sandra greets her first and then asks her, "Is there a specific place that you would like to be for our conversation today?"

Lisa says, "Yes. I have dreamt of my son several times this month, and most of the time, the locations are in the hospital, either a triage room or an in-patient room. Maybe we can talk in an in-patient room where I stayed for a long time with my son during his treatment."

"Sure. We can do that!" Sandra replies.

"OK. Let me generate the room!" Lisa says. She then selects the Tools icon from her left virtual hand and expands it, from where she selects *MemoryViz* (a generative AI tool that helps visualize a past place memory as a 3D environment in VR) > *Audio* (describe the place with audio input). When the animated microphone icon pops up, she starts her verbal description of an in-patient hospital room with details: "This is an in-patient room on the fourth floor of the hospital building. The room is spacious with a nurse station, a pediatric patient bed with guards, and a long blue couch by its big window…"

After she completes and confirms the extracted keyword prompts from her verbal description, the cabin room disappears, and a hospital room appears, surrounding Sandra and her. However, Lisa feels the AI-generated hospital room is somewhat different from what she has described from memory and mentions it to Sandra.

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Sandra asks Lisa, "Do you happen to have a photo of the past hospital room in your phone? If so, you may import it into *MemoryViz* to give it some reference for better accuracy."

"Yes, I do have some!" Lisa reopens the *MemoryViz* menu and finds the option, *Reference*. She selects it and follows the instructions to connect the VR headset to her smartphone media gallery and import a photo of the hospital room she took in the past. After a few seconds, the surrounding environment is refreshed based on the imported photo reference, and Lisa looks around and feels quite satisfied with the second attempt at image generation.

"This place looks familiar to me. Let's start our session now," Lisa speaks to Sandra.

"Great! I am sure you will have a lot of memories to share in this room, and I am ready to listen," Sandra responds.

4 Conclusion and Future Work

Meaning-centered grief therapies are promising interventions that can benefit bereaved parents in helping them live through a healthy grieving process after the loss of their children. To overcome the accessibility constraints of the in-person format of the interventions and the lack of feature support in remote format, we proposed a design solution of a networked, multi-user IVR therapeutic system with enhancement of text-to-image genAI models to support the core features of meaning-centered grief activities and simulate a real-world therapeutic experience. Although our current effort on project development is focused on the bereaved parents, the design concept and framework of the system development can be adapted and applied to other bereavement populations and other psychotherapies. Our next step of the project is to complete the project prototype development and use it for a pilot study with a few bereaved parents to further validate the feasibility of the design concept. A longitudinal and quantitative study is also in the planning to further investigate how different aspects of system designs of an IVR system may affect emotion processing and grieving experiences of the bereaved individuals.

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